

WilliamShen

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Software developer driven by creativity and innovation to deliver unique tech solutions

SKILLS

Technologies Java, React, HTML, CSS, JS, SQL, R, Python, C++, C, C#

Tools Git, MongoDB, AWS, Firebase, Bash, Powershell, Splunk

EXPERIENCE

SQL Server/Infrastructure DBA | *SunLife*

Sep 2025 - Dec 2025

- Automated mapping of **400+ CIS** standards to **Imperva** scans with **OpenPyXL** in **Python** for **MongoDB, MSSQL, PostgreSQL, Oracle, MySQL**, reducing manual compliance review time by **70%**
- Developed **Splunk** dashboards for **MSSQL, Guardium Data Encryption, Oracle Key Vault**, creating **30+** panels to monitor security operational issues (e.g., suspicious logins, high traffic, backup errors)
- Scripted **Active Directory** functional ID expiry monitoring in **PowerShell**, querying attributes via **SQL** for **600+** accounts across **DEV, UAT, STAGE, PROD** environments
- Researched and proposed **Governance & Compliance AI Agent** for Continuous Improvement, presenting feasibility, implementation, and impacts to senior leadership

App Developer & Code Sensei | *Code Ninjas*

Jan 2025 - Apr 2025

- Developed full-stack app using **Next.js, Tailwind CSS**, and **Firebase** to display student session statistics—implemented **CRUD** functions on admin page, and grid display on session display page
- Standardized formatting of human inputted times on **Excel** spreadsheet automatically using **regex** and **csv** in **Python**, to improve data entry time by **109%**
- Updated incomplete product information on **Shopify** using **Shopify API** for **400+** items automatically
- Taught various levels of game development (scripting, tilemap and sprite design, etc.) on platforms such as **Microsoft MakeArcade, Roblox Studio**, and **Unity**, to **70+** kids aged 5-12 every week

Robotics Camp Supervisor | *Young Engineers*

May 2024 - Aug 2024

- Managed team of **15+ counselors** and guided **100+ students** aged 4-10 in robotics projects
- Created and led daily lessons and activities on robotics and mechanics for all skill levels

PROJECTS

Communal Catalogue | *Personal*

Jan 2025 - May 2025

- Designed and developed a full-stack library management system using **React Vite** and **Firebase** empowering small communities to organize, track, and manage book checkouts easily
- Built a webcam ISBN scanner using **QuaggaJS**, allowing users to quickly checkout books by scanning barcodes, using **OpenLibrary API** to fetch book details
- Implemented role-based authentication system with **Firebase Auth** to manage admin access

Gendently | *HackTheNorth*

Sep 2024

- Designed and developed a fullstack web application leveraging technologies such as **React, Flask**, and **Scikit-learn** to predict gender of a given name using machine learning models
- Built a **React.js** frontend to capture user input, integrated with a **Flask** backend API to handle requests, displaying the predicted gender in real time
- Integrated **MongoDB** as the database for storing user inputs and predictions, efficiently managing data persistence and retrieval on the backend
- Trained gender prediction model using **scikit-learn** on a dataset with **150,000** names & genders

EDUCATION

University of Waterloo | *Candidate for Bachelor of Mathematics*

Sep 2023 - Present

- Major: Computational Math | Minor: Combinatorics & Optimization
- President's Scholarship